

TDD Green Phase - Make Tests Pass Quickly

Write the minimal code necessary to satisfy GitHub issue requirements and make failing tests pass. Resist the urge to write more than required.

GitHub Issue Integration

Issue-Driven Implementation

- **Reference issue context** - Keep GitHub issue requirements in focus during implementation
- **Validate against acceptance criteria** - Ensure implementation meets issue definition of done
- **Track progress** - Update issue with implementation progress and blockers
- **Stay in scope** - Implement only what's required by current issue, avoid scope creep

Implementation Boundaries

- **Issue scope only** - Don't implement features not mentioned in the current issue
- **Future-proofing later** - Defer enhancements mentioned in issue comments for future iterations
- **Minimum viable solution** - Focus on core requirements from issue description

Core Principles

Minimal Implementation

- **Just enough code** - Implement only what's needed to satisfy issue requirements and make tests pass
- **Fake it till you make it** - Start with hard-coded returns based on issue examples, then generalise
- **Obvious implementation** - When the solution is clear from issue, implement it directly
- **Triangulation** - Add more tests based on issue scenarios to force generalisation

Speed Over Perfection

- **Green bar quickly** - Prioritise making tests pass over code quality

- **Ignore code smells temporarily** - Duplication and poor design will be addressed in refactor phase
- **Simple solutions first** - Choose the most straightforward implementation path from issue context
- **Defer complexity** - Don't anticipate requirements beyond current issue scope

Implementation Strategies (Polyglot)

- **Start with constants** - Return hard-coded values from issue examples initially
- **Progress to conditionals** - Add if/else logic as more issue scenarios are tested
- **Extract to methods/functions** - Create simple helpers when duplication emerges
- **Use basic collections** - Simple arrays, lists, or maps over complex data structures

Execution Guidelines

1. **Review issue requirements** - Confirm implementation aligns with GitHub issue acceptance criteria
2. **Run the failing test** - Confirm exactly what needs to be implemented
3. **Confirm your plan with the user** - Ensure understanding of requirements and edge cases. NEVER start making changes without user confirmation
4. **Write minimal code** - Add just enough to satisfy issue requirements and make test pass
5. **Run all tests** - Ensure new code doesn't break existing functionality
6. **Do not modify the test** - Ideally the test should not need to change in the Green phase.
7. **Update issue progress** - Comment on implementation status if needed

Green Phase Checklist

- ☐ Implementation aligns with GitHub issue requirements
- ☐ All tests are passing (green bar)
- ☐ No more code written than necessary for issue scope
- ☐ Existing tests remain unbroken
- ☐ Implementation is simple and direct
- ☐ Issue acceptance criteria satisfied
- ☐ Ready for refactoring phase